

System requirements

Last Updated: 2026-03-28

The system requirements for installing IBM Fusion software.

The IBM Fusion operators and your workloads run on OpenShift® Container Platform that in turn uses 64-bit Intel, AMD x86, Linux on IBM Z, or Power hardware architecture. For more information about Fusion data services on the different deployment platforms of IBM Fusion, see [IBM Fusion services and platform support matrix](#).

The following table lists the system requirements for IBM Fusion software:

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Note: Though there is no minimum number of nodes requirement, the following table provides data based on three control nodes and three compute nodes cluster. However, sizing changes based on the number of nodes for components with "Per node".

Fusion Data Foundation can be deployed on Infra or worker nodes, whereas all other IBM Fusion services including Base must be deployed on worker nodes.

Component	vCPUs	Memory	Storage
IBM Fusion Base	Request CPU: 4 Limit CPU: 10	Request memory: 4 GiB Limit memory: 13 GiB	Overall storage is a minimum of 500 MB.
Fusion Data Foundation			
Fusion Data Foundation storage	Internal mode (Local and dynamic) Request CPU: 22+2* (total count of Object Storage Daemon)	Internal mode (Local and dynamic) Request memory: 84+5* (total count of Object Storage Daemon) GiB Limit memory:	Minimum three storage Nodes For more information about Fusion Data Foundation, see Important considerations when deploying Red Hat OpenShift Data Foundation .

Component	vCPUs	Memory	Storage
	Limit CPU: 26+2* (total count of Object Storage Daemon)	88+5* (total count of Object Storage Daemon) GiB	For IBM Power Systems: Internal 3 storage devices, each with additional 500 GB of disk
	External mode Request CPU: 6	External mode Request memory:16 GiB	
	Limit CPU: 6	Limit memory: 16 GiB	
	For IBM Power Systems: Internal mode	For IBM Power Systems: Internal mode 192 GiB memory	
	48 CPU (logical)	External mode 48 GiB memory	
	External mode 24 CPU (logical)		
Fusion Data Foundation MCG	Core CPU: 1	Core Memory: 4 GiB	
	Database CPU: 1	Database Memory: 4 GiB	NA
	Endpoint CPU: 1	Endpoint Memory: 2 GiB	

Component	vCPUs	Memory	Storage
Global Data Platform	Per node	Per node	For more information about IBM Spectrum Scale Container Native Storage Access, see Hardware requirements for IBM Storage Scale Container Native Storage Access .
	Request CPU: 10% (total vCPU of the node)	Request memory: 10% (total memory of the node)	
	Cluster	Cluster	
	Request CPU: 3	Request memory: 14	
	Limit CPU: 14	Limit memory: 35	
Backup & Restore			
Backup & Restore Base Hub install requirements (assuming 3 node cluster)			– Request: Ephemeral Storage: 3.7 GiB
	– Request CPU: 4.94	– Request Memory: 10.76 GiB	PVC Storage: 200 GiB
	– Limit CPU: 27.58	– Limit Memory: 34.35 GiB	– Limit: Ephemeral Storage: 7 GiB
			PVC Storage: NA
Backup & Restore Base Spoke install requirements (3 node cluster). For more information scaling the hub, see Backup & restore hub performance and scaling .	– Request CPU: 2.3	– Request Memory: 2.1 GiB	– Request: Ephemeral Storage: 950 MiB
	– Limit CPU: 10.2	– Limit Memory: 11 GiB	PVC Storage: 10 GiB
			– Limit:

Component	vCPUs	Memory	Storage
			Ephemeral Storage: 1650 MiB PVC Storage: NA
Each additional node for Backup & Restore (Beyond the first three) Applicable for both Hub and Spoke)	– Request CPU: 0.2 – Limit CPU: 0.4	– Request Memory: 150 MiB – Limit Memory: 800 MiB	– Request Ephemeral Storage: 50 MiB – Limit Ephemeral Storage: 500 MiB
Per Job Additional Requirements: Data Mover (1 per PVC, max 5 per node)	Request CPU: 2 Limit CPU: 4	Request Memory: 2 GiB Limit Memory: 10 GiB	Request Ephemeral Storage: 5 GiB Limit Ephemeral Storage: 5 GiB
Per Job Additional Requirements: Legacy Restic Data Mover (1 per PVC, max 3 per node)	Request CPU: 0.1 Limit CPU: 2	Request Memory: 4,000 MiB Limit Memory: 15,000 MiB	Request Ephemeral Storage: 400 MiB Limit Ephemeral Storage: 2,000 MiB
Per Job Additional Requirements: Maintenance (after every job and once per day per application/policy combination)	Request CPU: 0.5 Limit CPU: 4	Request Memory: 500 MiB Limit Memory: 4 GiB	Request Ephemeral Storage: 5 GiB Limit Ephemeral Storage: 5 GiB
IBM Data Cataloging	Request CPU: 21	Request memory: 27	120 GB Minimum

Component	vCPUs	Memory	Storage
	Limit CPU: 78	GiB Limit memory: 126 GiB	For more information about IBM Data Cataloging, see Installing IBM Data Cataloging .
Content-Aware Storage (CAS)			
Content-Aware Storage with less than 12 TB of ingested data Note: For Content-Aware Storage with more than 12 TB of Ingested Data, contact IBM Sales representatives for your intended workloads.	Request: 9 Limit: 25	Request: 22 GiB Limit: 24 GiB	The minimum required filesystem size is 550 GB. For more information about IBM Spectrum Scale Container Native Storage Access, see Hardware requirements for IBM Storage Scale Container Native Storage Access .
nv-ingest Requirements	Request: 30 Limit: 60	Request: 90 GiB Limit: 290 GiB	100 GB
docling_multimodal - Additional requirements	Request: 4.5 Limit: 21	Request: 13 GiB Limit: 44 GiB	100 GB
Minimum requirements for each additional Collection	Request: 1 Limit: 2	Request: 5 GiB Limit: 5 GiB	1 GB

GPU resources

Configuration	GPU worker nodes	Non-GPU worker nodes	Production with Multi-Instance GPU (MIG)* enabled	Production without Multi-Instance GPU (MIG) enabled
Content-Aware Storage with less than 12 TB of ingested data	1	2	2 or more MIG Capable NVIDIA GPUs (A100 80GB, H100, H200, RTX PRO 6000)	1 NVIDIA GPU for each NVIDIA NIM service** configured (L40S, A100 H100, H200, RTX PRO 6000)

* For more information about MIG, see [Configuring a Multi-Instance GPU \(MIG\) with CAS](#).

** For more information about the NVIDIA NIM services, see [Select NVIDIA Inference Microservices \(NIMs\)](#).

The following table lists the component versions:

Component	Versions
On-premises VMware	Version 7.0
On-premises IBM z/OS Container Extensions (IBM zCX)	All the zCX hypervisor versions on which OpenShift Container Platform is supported.
Red Hat® OpenShift Container Platform	4.16, 4.17, 4.18, 4.19, and 4.20. <i>Note:</i> The console plugin does not support the OpenShift Container Platform versions of Fusion Backup & Restore service. For more information, see Guidance for Fusion Native Console plugin OCP version.
Red Hat OpenShift Kubernetes Engine (OKE)	You can use OKE as an alternative for OpenShift Container Platform. The supported version of OKE is same as OpenShift Container Platform.
Red Hat OpenShift Virtualization Engine (OVE)	IBM Fusion provides support for OVE as an alternative for OpenShift Container Platform. The supported version of OVE is same as OpenShift Container Platform. For more information about the limitations and considerations of using OVE, see OVE considerations .
Storage	Fusion Data Foundation 4.16, 4.17, 4.18, 4.19, and 4.20

Component	Versions
	Note: For deployment platforms and supported services, see Support matrix for IBM Fusion versions. IBM Storage Scale Note: IBM Fusion supports IBM Storage Scale 5.2.3.x. Remote mount of IBM Storage Scale storage: – IBM Storage Scale remote storage cluster release 5.1.9.0 or higher. – The IBM Fusion cluster accesses the storage owned by an IBM Storage Scale storage cluster by using a remote mount. The IBM Storage Scale file system version of the owning storage cluster cannot be newer than version 37.00. To determine the version of your IBM Storage Scale cluster, run the mmdia --version command on the IBM Storage Scale cluster. To determine the version of your IBM Storage Scale file system, run the mmlsfs all -V command. For further information, including installation/upgrade instructions for the remote IBM Storage Scale cluster, see IBM Storage Scale documentation.
Backup & Restore	IBM Fusion supports backup and recovery operations on Red Hat OpenShift Container Platform environment with any storage provider that implemented the Container Storage Interface (CSI) driver and meets the following prerequisites: – The Container Storage Interface driver must be v1 or higher (no support for <i>alpha</i> or <i>beta</i> versions) – The Container Storage Interface driver must support Volume Snapshot and Restore. – StorageClass must be created with <code>volumeBindingMode: Immediate</code> or <code>volumeBindingMode: WaitForFirstConsumer</code> and <code>allowVolumeExpansion: true</code>. – PersistentVolume (PV) must be created with <code>volumeMode: Filesystem</code> or <code>volumeMode: Block</code> Note: It is also recommended that the VolumeSnapshotClass deletion policy is configured to delete snapshots after the

Component	Versions
	corresponding VolumeSnapshotContent is deleted (deletionPolicy: Delete). i Note: For backups of PVCs provisioned on IBM Storage Scale, snapshots are be created only from independent fileset-based persistent volume claims (PVCs). PVCs that are based on lightweight directories and dependent file sets are not supported.

Parent topic:

→ [Prerequisites](#)